

VLE 2 @ Linux commands



TEAM 10

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Scenario 1 — Student Records Folder Setup

List files in current directory

```
ls
```

Move into Sem-6 folder

```
cd Sem-6
```

Create empty file

```
touch records.txt
```

View file content (should be empty)

```
cat records.txt
```

Scenario 2 — Append & Redirection

Append data

```
echo"RegNo: 1023 | Name: Arjun | Grade: A" >> records.txt
```

Overwrite file

```
echo"RegNo: 1024 | Name: Ravi | Grade: B" > records.txt
```

Save file list to file

```
ls > filelist.txt
```

Scenario 3 — STDIN / STDOUT / STDERR

Definitions

- **STDIN (0)** → input from keyboard
- **STDOUT (1)** → normal output to screen
- **STDERR (2)** → error messages

Example commands

STDIN

```
cat > input.txt
```

(type text → Ctrl+D)

STDOUT

```
echo "Hello World"
```

STDERR

```
ls wrongfile 2> error.txt
```

Scenario 4 — grep searching

Lines containing "Grade"

```
grep "Grade" records.txt
```

Lines NOT containing "A"

```
grep -v "A" records.txt
```

Pipe search (Grade + B)

```
grep "Grade" records.txt | grep "B"
```

Scenario 5 — cut command

Extract name (2nd word)

```
cut -d" " -f4 records.txt
```

("Ravi" or "Arjun" depending on data)

First 5 characters

```
cut -c1-5 records.txt
```

Scenario 6 — awk

Print Grade field

```
awk '{print $NF}' records.txt
```

(NF = last field → Grade)

OR

```
awk '{print $6}' records.txt
```

Count lines

```
awk 'END {print NR}' records.txt
```

Scenario 7 — Final Pipeline (MOST IMPORTANT ★)

Requirement:

- ✓ filter "A"
- ✓ extract name
- ✓ first 3 characters

Final command:

```
grep"A" records.txt |cut -d" " -f4 |cut -c1-3
```

OR using awk:

```
grep"A" records.txt | awk'{print substr($4,1,3)}'
```